

Robots Lending a Helping Hand

Robots	How do they help
Exoskeletons like Raytheon's Sarcos and Cyberdyne's HAL (Hybrid Assistive Limb)	Reduce effort
Advanced wheeled and legged robots with autonomous navigation capabilities such as Black-Is Land Shark, Lockheed Martin's SMSS and Boston Dynamics' BigDog	Locomotion on rough terrains
Sandia's Gemini Scout, tele-surgery robots	Evacuation and treatment of wounded
Autonomous aerial and underwater vehicles, and semi-autonomous ground robots	Remote surveillance

robots are very much related to the advancement in robotics, vision and artificial intelligence technologies. It may take one or two decades for robots to search, identify and track enemy forces and make decisions to call in and coordinate strikes from unmanned land vehicles, pilot-less aircrafts and underwater vehicles in a fully autonomous way. Until then, these systems would assist soldiers in warfare to lighten their loads, keep them better informed and perform strikes through remote operation," says Jayakrishnan.

However, there is a new technology that takes a middle path—exoskeletons. Here, a human inside the robot combines the intelligence of the person with the strength of the exoskeleton. Raytheon's Sarcos XOS exoskeleton arms and legs are one of the most well-known models for use in the military. The design weighs 68 kg and allows the wearer to lift 90 kg without any special effort. XOS 2 is a recent model featuring more fluid movement, increased power output and decreased power input.

Berkeley Bionics/Lockheed Martin HULC (human universal load carrier) legs are a key contender to XOS. The carrier weighs only 24 kg and allows the user to carry up to 91 kg on a backpack attached to the exoskeleton independent of the user.

Surveillance bots gaining popularity

Surveillance from a defence perspective mainly involves operating from a stable and reliable platform, camouflaging and information gathering with

the help of sensors and vision systems.

Jayakrishnan explains that stealth or low-observable technology has been used for long for being invisible from radars. Apart from traditional camouflaging patterns, digital camouflaging techniques are also widely used to

dress-up battlefield vehicles and equipment nowadays. New technologies such as adaptive camouflaging have been developed using nanotechnology. For UGVs, the vision system or smart code can monitor an area for intruders and alert commanders when an event is detected.

Today, there are many advanced UGVs, UAVs and UAVs being used in surveillance platforms. Of late, drones are highly discussed in the news and have actually served to bring defence robotics into the limelight. The Stalker (a product of Lockheed Martin) is a stealthy aerial drone that was used by US Special Forces in Afghanistan. It weighs 6 kg, has a three-metre wingspan and is launched by hand. These drones are designed to be quiet with a 'hush drive' that combines a silenced electric motor and a special propeller. The Stalker is said to be inaudible beyond 80 metres and can fly at night with the help of infrared sensors and low-light cameras.

Larger autonomous aerial vehicles that can send data or images in real time to base stations are also used for surveillance. Elbit Systems' latest large UAV offering Hermes 900 extends its predecessor Hermes 450's performance

to offer longer endurance, higher operational ceiling, larger payload capacity and all-weather functionality. Add to this a silent engine, innovative electronics and avionics, electro-optic systems, laser designators and sensor technology, 36-hour endurance, speed of 222.23 kmph (120 knots) and a payload capacity of 300 kg—what you have is a surveillance plus payload carrier machine.

"We have seen and heard of many robots designed and inspired by functionalities of the structures of many living organisms such as birds, fishes and reptiles. The Snakebot is one such robot used for surveillance. The 1.82-metre (six feet) Snakebot developed by Biorobotics and Biomechanics Lab of the Technion-Israel Institute of Technology consists of polymer segments connected by flexible joints, and is powered by electric motors. Movement control relies on a software that determines the best mode of travel (wriggling, rolling or corkscrewing) for each situation. Snakebot can also rise up to climb stairs and other vertical obstacles. Even NASA is developing Snakebots to explore space," says Shekhar.

Harvard University's 60-milligram robot with a three-centimetre wingspan and Festo's SmartBird robot are other examples of nature-inspired surveillance bots.

Against all odds

"Robots in the defence sector are designed for harsh weather conditions and their performance is always compared with their human counterparts in such conditions. These don't need life-support systems to survive harsh conditions and can go for long hours or days on single-mission deployment. Even if a robot is lost in action, another one can be sent. There is no need for a rescue operation, which in itself is a huge saving in the operating cost and precious lives," says Fahad Azad, managing partner, RoboSoft Systems.

From this perspective, Azad appreciates the work of Boston Dynam-

Like any other standard application, robotics, when applied to defence, optimises speed, accuracy, tirelessness and capability